

MACHINE LEARNING USING TREES

Breast Cancer Wisconsin Diagnostic Dataset

Course: CS 4372

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Assignment: Machine Learning Using Trees

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1. Introduction

Machine learning classification methods can be used to identify patterns in medical measurements and predict whether a tumor is benign or malignant. In this project, four tree-based machine learning methods are developed and compared using the Breast Cancer Wisconsin Diagnostic dataset.

The four models evaluated are:

1. Decision Tree Classifier
2. Random Forest Classifier
3. AdaBoost Classifier
4. XGBoost Classifier

The objective is to determine which model provides the best classification performance while examining the data characteristics, preprocessing requirements, feature relationships, model parameters, and evaluation metrics.

Because this is a binary classification problem, accuracy alone is not sufficient to evaluate model performance. Therefore, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrices are also examined.

2. Dataset Description

The Breast Cancer Wisconsin Diagnostic dataset contains measurements computed from digitized images of fine needle aspirate samples of breast masses. The observations describe characteristics of cell nuclei.

The dataset contains **569 observations and 33 columns** in its original form. The target variable is diagnosis, which indicates whether a tumor is:

- **B** — Benign
- **M** — Malignant

The dataset contains 30 numerical predictor variables. These measurements describe characteristics including:

- radius
- texture
- perimeter
- area
- smoothness
- compactness
- concavity
- concave points
- symmetry
- fractal dimension

Each measurement is represented using mean, standard error, and worst-case measurements.

The dataset was loaded directly from a publicly accessible UTD Box URL rather than being submitted to Canvas.

Public Dataset URL : [UTD Box dataset](#)

3. Data Preprocessing

3.1 Loading the Dataset

The dataset was loaded into a Pandas DataFrame using the public UTD Box URL. The original dataset contained 569 observations and 33 columns.

The first few observations were examined to understand the structure and contents of the dataset.

3.2 Missing Values and Data Consistency

The dataset was examined for missing values.

All meaningful variables contained zero missing values. The only column containing missing values was Unnamed: 32, which contained missing values for all 569 observations. Therefore, this column was completely empty and was removed.

The id variable was also removed because it is an identifier and does not represent a meaningful predictive characteristic.

After removing these two columns, the dataset contained:

569 observations and 31 columns, consisting of 30 numerical predictors and one target variable. No duplicates were found.

4. Target Variable

The original diagnosis variable contained two categorical values:

- B = Benign: **357 observations**
- M = Malignant: **212 observations**

This corresponds to approximately:

- **62.74% benign**
- **37.26% malignant**

The target was converted into numerical form:

- 0 = Benign
- 1 = Malignant

This conversion allows the classification algorithms to process the target variable numerically.

The classes are somewhat imbalanced because benign observations are more common than malignant observations. However, the minority class still represents more than one-third of the dataset, so stratified sampling and metrics such as recall and F1-score are important.

5. Summary Statistics

Descriptive statistics were calculated for the dataset.

The predictors have substantially different numerical scales. For example, variables such as `area_mean` and `area_worst` have values in the hundreds or thousands, whereas variables such as `smoothness_mean` and `fractal_dimension_mean` have much smaller values.

This difference in scale motivated the use of standardization and normalization as part of preprocessing.

6. Distribution of Attributes

The distributions of the numerical variables were examined using histograms and skewness statistics.

The attributes are not all normally distributed. Several variables show substantial positive skewness. For example:

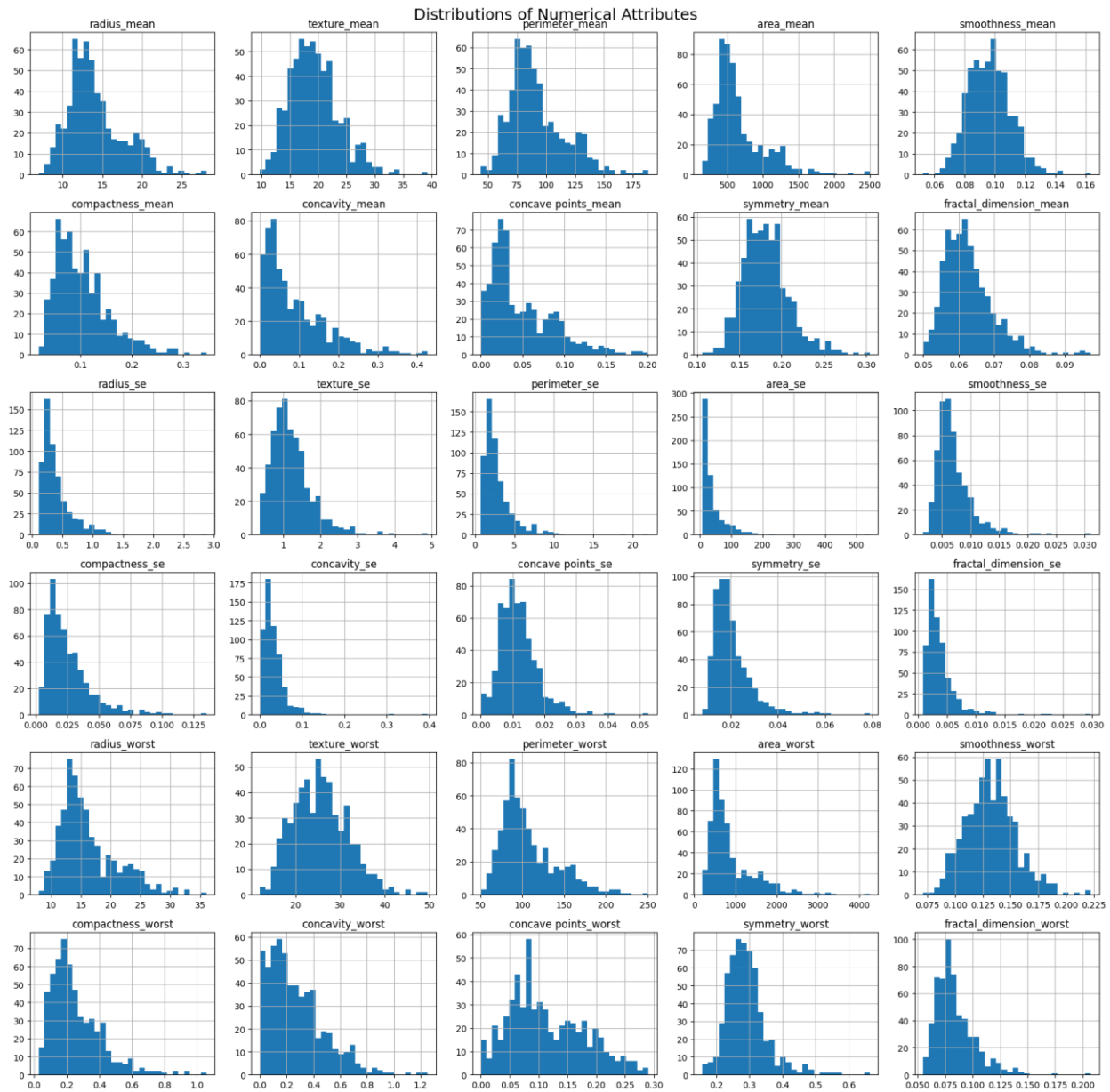
- `area_se`: skewness ≈ 5.45
- `concavity_se`: skewness ≈ 5.11

- fractal_dimension_se: skewness ≈ 3.92
- perimeter_se: skewness ≈ 3.44
- radius_se: skewness ≈ 3.09

Other variables have considerably smaller skewness values.

Interpretation

The attributes are not normally distributed. Several of the measurements, particularly some standard-error and area-related variables, show strong positive skewness. This is reasonable because the attributes represent biological measurements with varying ranges and natural variability. Tree-based models do not require the predictors to follow a normal distribution, so the observed non-normality does not prevent the models from being used.



7. Standardization and Normalization

Both standardization and normalization were performed.

Standardization was performed using StandardScaler, which transforms each feature based on its mean and standard deviation.

Normalization was performed using MinMaxScaler, which scales the features to a common range.

Although tree-based models generally do not require feature scaling because tree splits depend on feature thresholds rather than distances, scaling was performed because it was explicitly required by the assignment.

For the final models, the selected training features were standardized using parameters learned only from the training data. The same transformation was then applied to the test data.

8. Correlation Analysis

A correlation matrix was calculated to examine relationships among the numerical predictors and the target variable.

The correlation analysis showed that several measurements related to tumor size and shape have strong relationships with the diagnosis variable.

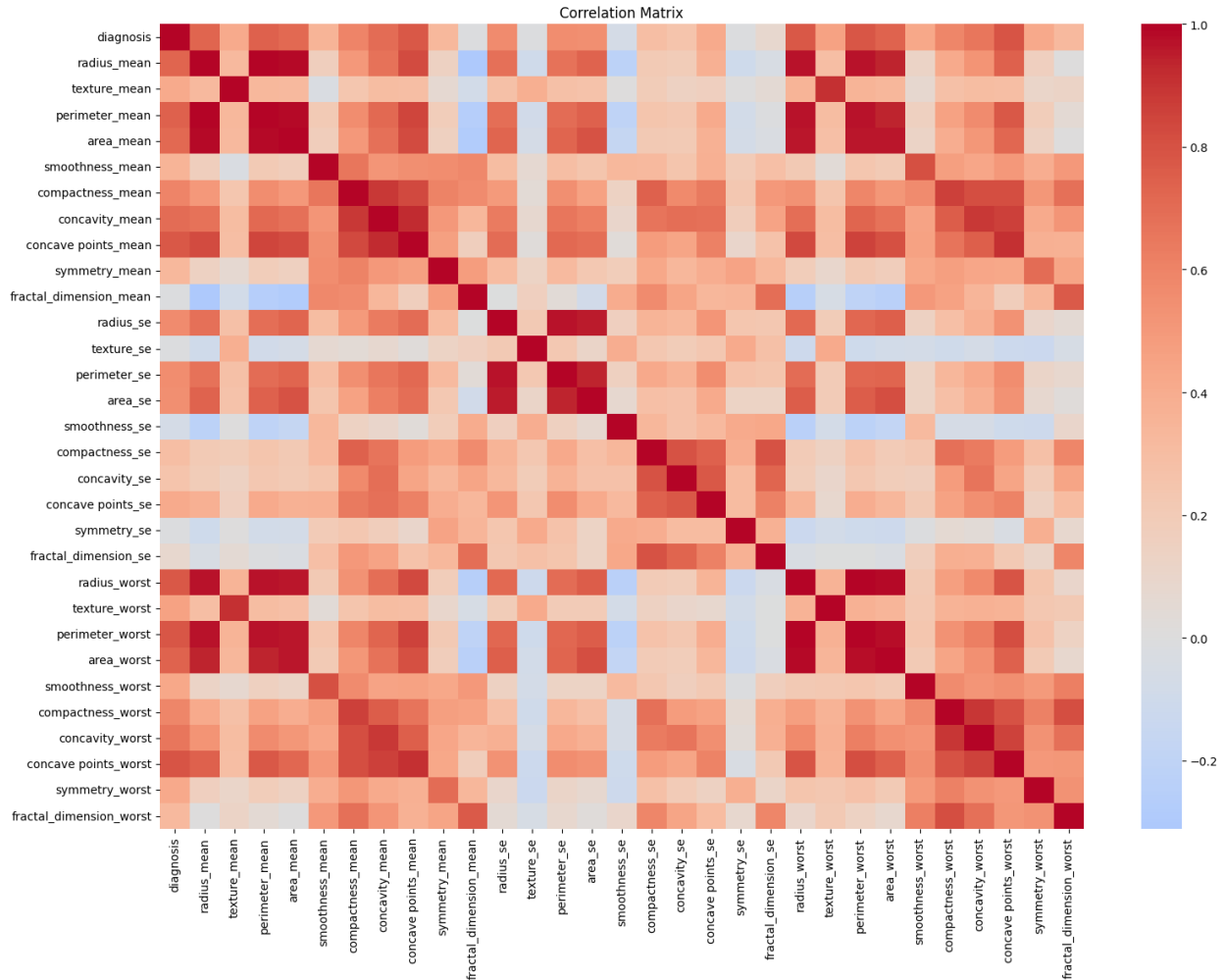
The strongest predictors selected using the training data were:

1. concave points_worst
2. perimeter_worst
3. radius_worst
4. concave points_mean
5. perimeter_mean
6. area_worst
7. radius_mean
8. area_mean

These eight variables were used for the final model construction.

Interpretation

The correlation analysis indicates that measurements related to tumor radius, perimeter, area, and concave points are particularly informative for distinguishing between benign and malignant tumors. The worst measurements and concave-point measurements appear especially useful. Because many of these variables describe related physical characteristics, some predictors are also strongly correlated with each other.



9. Feature Selection

The assignment requires selecting important attributes rather than blindly using all available predictors.

The data was divided into training and testing sets before the final feature selection. Feature correlations with the target were calculated using only the training data to reduce the possibility of test-data leakage.

The eight selected predictors were:

Rank	Feature
1	concave points_worst

2	perimeter_worst
3	radius_worst
4	concave
4	points_mean
5	perimeter_mean
6	area_worst
7	radius_mean
8	area_mean

These variables were used for the four tree-based models.

10. Train-Test Split

The dataset was divided into training and testing sets using an 80/20 split.

A stratified split was used to preserve the proportion of benign and malignant observations in both sets.

The resulting datasets were:

- Training set: **455 observations**
- Testing set: **114 observations**

The random state was set to 42 to provide reproducible results

11. Model Construction

Four tree-based classification algorithms were constructed:

11.1 Decision Tree

A Decision Tree creates a sequence of decision rules that divide observations into increasingly homogeneous groups.

11.2 Random Forest

Random Forest combines multiple decision trees and uses their collective predictions. This ensemble approach can reduce the variance associated with a single decision tree.

11.3 AdaBoost

AdaBoost builds a sequence of weak learners, giving additional emphasis to observations that were incorrectly classified by previous learners.

11.4 XGBoost

XGBoost is a gradient-boosting approach that sequentially builds trees to improve the predictions of previous trees. It also incorporates regularization to help control model complexity.

12. Hyperparameter Tuning

GridSearchCV with **5-fold Stratified Cross-Validation** was used to tune the models.

F1-score was used as the GridSearchCV scoring criterion because the classification problem contains somewhat imbalanced classes and F1-score provides a balance between precision and recall.

The best parameters found by the completed notebook were:

Model	Best Parameters
Decision Tree	Criterion = entropy, Max Depth = 8, Min Samples Leaf = 5, Min Samples Split = 2
Random Forest	Max Depth = None, Max Features = sqrt, Min Samples Leaf = 2, N Estimators = 100
AdaBoost	Learning Rate = 0.5, N Estimators = 200
XGBoost	Colsample Bytree = 1.0, Learning Rate = 0.1, Max Depth = 4, N Estimators = 100, Subsample = 1.0

These parameters were selected based on cross-validation performance.

13. Model Evaluation

The models were evaluated using:

- Accuracy

- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC
- Confusion matrix

Accuracy measures the overall proportion of correct classifications.

Precision measures the proportion of predicted malignant cases that were actually malignant.

Recall measures the proportion of actual malignant cases that were correctly identified.

F1-score is the harmonic mean of precision and recall.

ROC-AUC measures the model's ability to distinguish between the two classes across classification thresholds.

PR-AUC summarizes the precision-recall relationship and is particularly useful when class distributions are not perfectly balanced.

14. Model Comparison

The final test-set results were:

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Decision Tree	93.86%	97.30%	85.71%	91.14%	99.52%	98.99%
Random Forest	95.61%	100.00%	88.10%	93.67%	99.04%	98.60%
AdaBoost	94.74%	95.00%	90.48%	92.68%	99.22%	98.79%
XGBoost	94.74%	100.00%	85.71%	92.31%	99.21%	98.86%

These values come directly from the final notebook results

15. Confusion Matrix Analysis

The confusion matrices show how each model classified the 114 test observations.

Decision Tree

- True negatives: 71
- False positives: 1
- False negatives: 6
- True positives: 36

The Decision Tree correctly identified most benign observations but missed six malignant cases.

Random Forest

- True negatives: 72
- False positives: 0
- False negatives: 5
- True positives: 37

Random Forest produced no false positives and correctly classified 37 of the 42 malignant observations.

AdaBoost

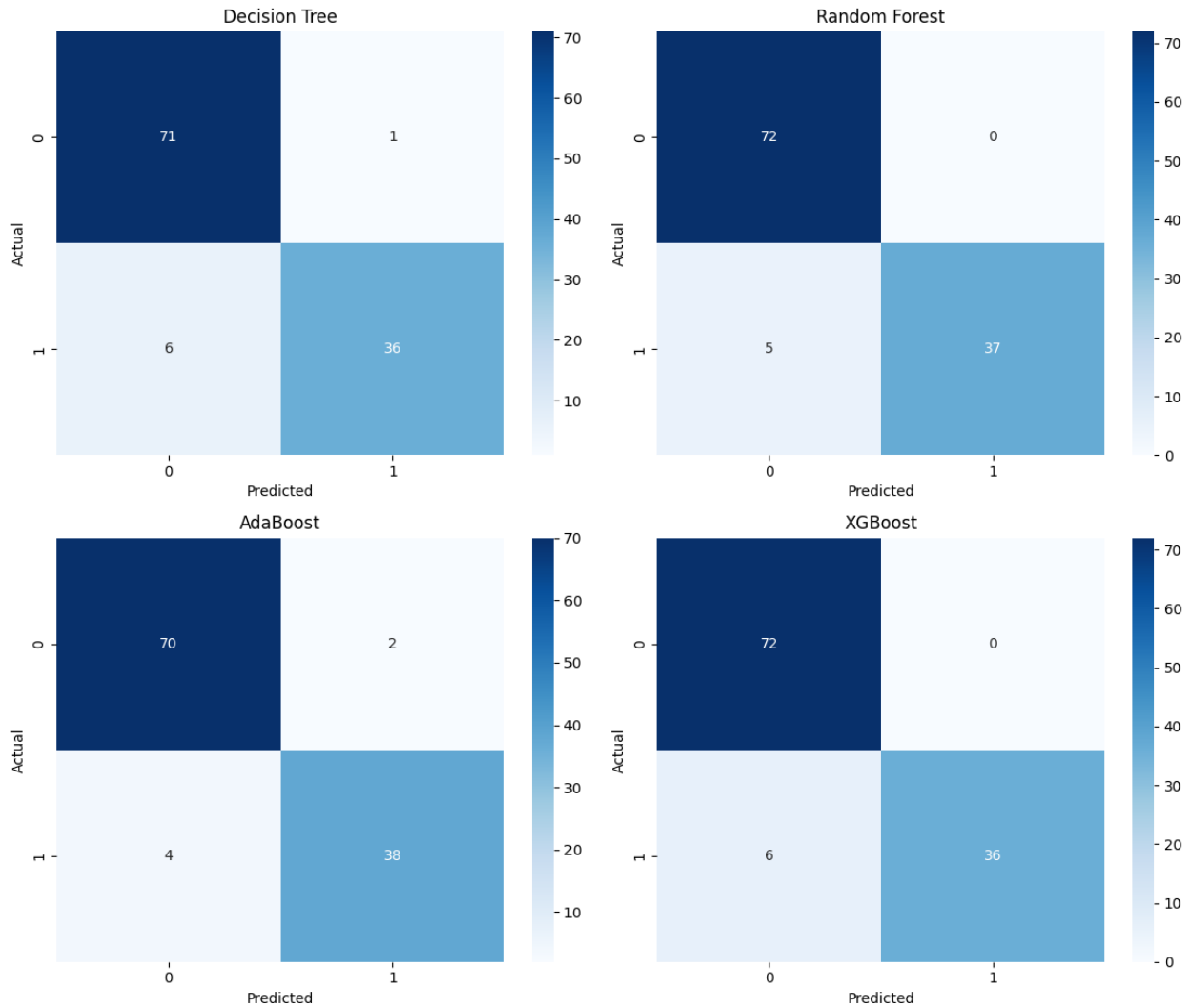
- True negatives: 70
- False positives: 2
- False negatives: 4
- True positives: 38

AdaBoost detected the largest number of malignant cases, with 38 true positives. However, it also produced two false positives.

XGBoost

- True negatives: 72
- False positives: 0
- False negatives: 6
- True positives: 36

XGBoost produced no false positives but missed six malignant observations.



17. Precision and Recall Analysis

The malignant class is particularly important in this problem because incorrectly classifying a malignant tumor as benign represents a false negative.

AdaBoost achieved the highest malignant recall:

90.48%

Random Forest achieved:

88.10%

Decision Tree and XGBoost achieved:

85.71%

Therefore, AdaBoost was the strongest model specifically in terms of detecting malignant observations.

However, Random Forest achieved 100% precision for malignant predictions, meaning that every observation predicted as malignant by Random Forest was actually malignant in the test set.

This demonstrates why multiple metrics are important. No single metric completely describes the behavior of these models.

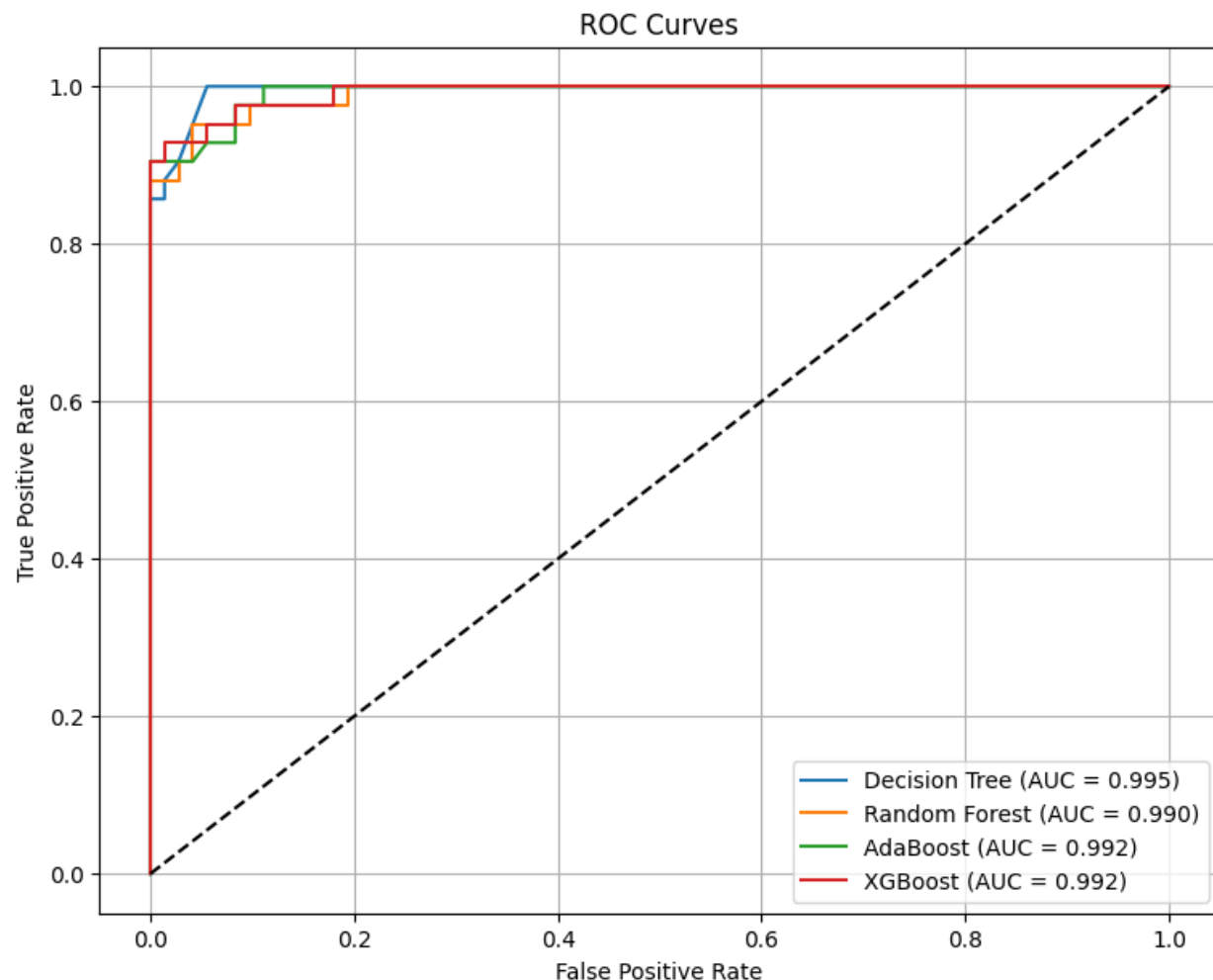
17. ROC Curve Analysis

The ROC curves showed that all four models performed very well.

The ROC-AUC values were:

Model	ROC-AUC
Decision Tree	0.9952
Random Forest	0.9904
AdaBoost	0.9922
XGBoost	0.9921

The Decision Tree had the highest ROC-AUC, although the difference among the models was very small.



Interpretation

All four models demonstrated excellent discriminatory ability, with ROC-AUC values above 0.99. This indicates that the models were highly effective at ranking malignant and benign observations according to their predicted probabilities.

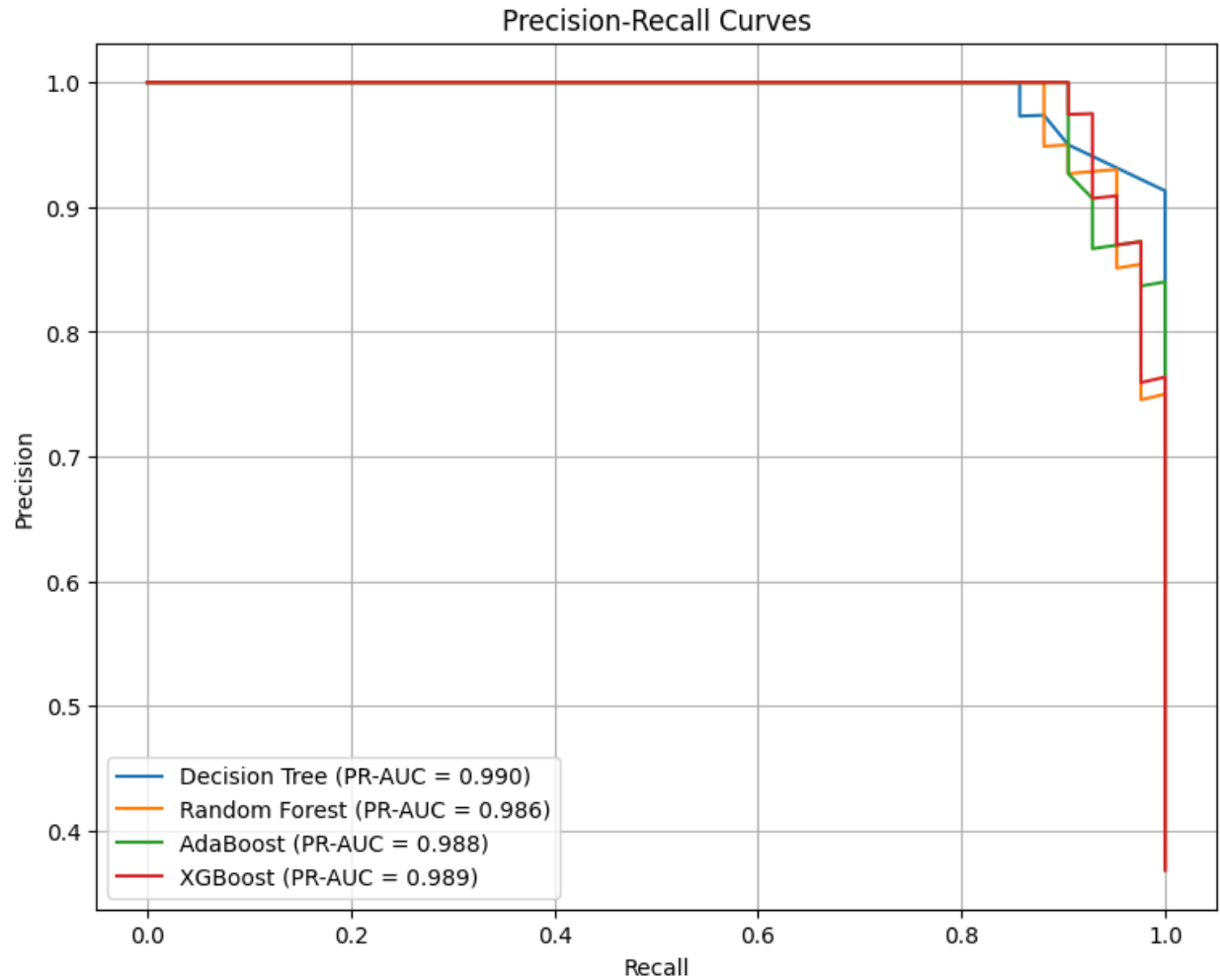
18. Precision-Recall Curve Analysis

The PR-AUC values were:

Model	PR-AUC
Decision Tree	0.9899
Random Forest	0.9860

AdaBoost	0.9879
XGBoost	0.9886

The Decision Tree achieved the highest PR-AUC, followed closely by XGBoost and AdaBoost.



Interpretation

The Precision-Recall results indicate strong performance for all four models. The high PR-AUC values show that the models maintained strong precision and recall across different classification thresholds.

19. Feature Importance

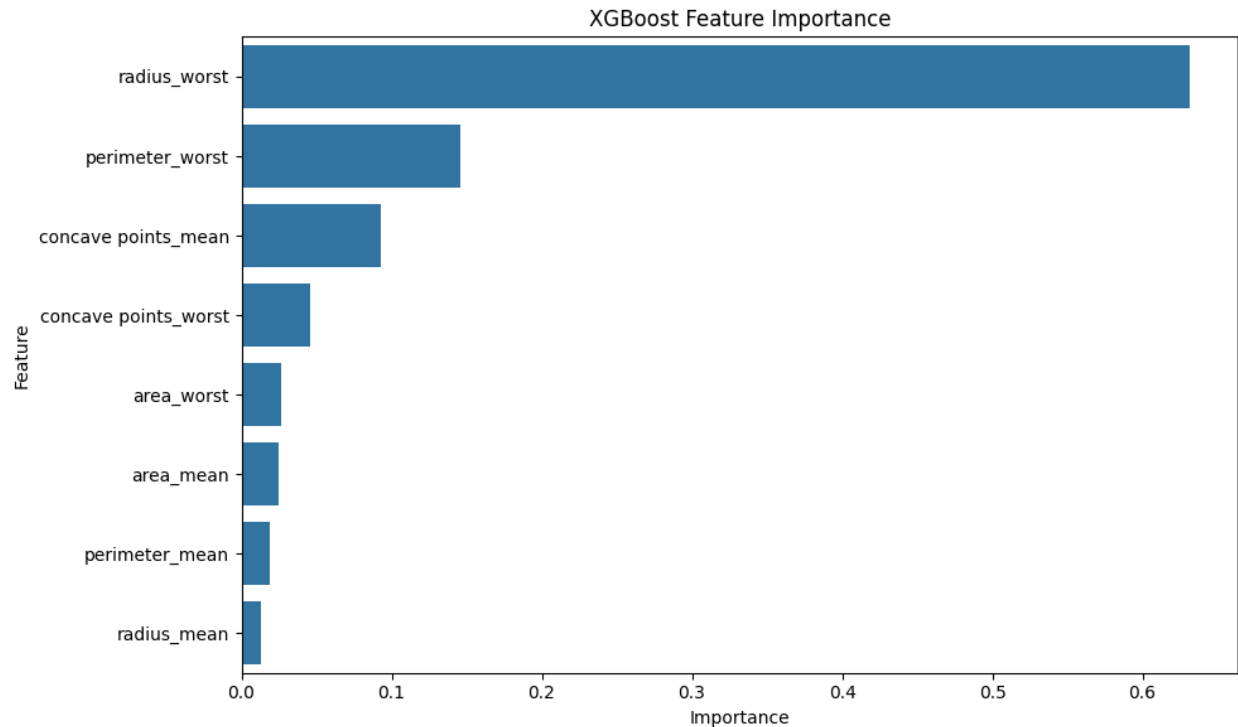
XGBoost feature importance was examined to determine which of the selected variables contributed most strongly to the model.

The feature importance values were:

Feature	Importance
radius_worst	0.6318
perimeter_worst	0.1459
concave	0.0931
points_mean	0.0455
concave	0.0455
points_worst	0.0266
area_worst	0.0250
area_mean	0.0186
perimeter_mean	0.0134
radius_mean	

The most important feature was **radius_worst**, with an importance of approximately **0.632**. **perimeter_worst** was the second most important feature at approximately **0.146**.

This indicates that the worst-case radius and perimeter measurements provided substantial predictive information for the XGBoost model.



20. Best Model

Based on the overall test-set performance, **Random Forest is the best overall model** in this experiment.

Random Forest achieved:

- **95.61% accuracy**
- **100% precision**
- **88.10% recall**
- **93.67% F1-score**
- **99.04% ROC-AUC**
- **98.60% PR-AUC**

It also produced zero false positives.

Although AdaBoost achieved slightly higher recall for malignant observations, Random Forest achieved the highest accuracy and F1-score and tied XGBoost for the highest precision.

Therefore, Random Forest provides the strongest overall balance among the evaluation metrics in this particular experiment.

21. Why Random Forest Performed Best

Random Forest combines the predictions of multiple decision trees rather than relying on one tree. This ensemble structure can reduce the variance of individual trees and improve generalization.

In this dataset, Random Forest achieved the highest overall accuracy and F1-score. It also produced no false positives and correctly identified 37 of the 42 malignant cases.

The individual Decision Tree performed slightly worse, which is consistent with the possibility of greater variance when relying on a single tree.

AdaBoost had the highest malignant recall, but its two false positives reduced its overall precision.

XGBoost performed strongly, but in this particular train-test split its malignant recall was lower than AdaBoost's and its accuracy was lower than Random Forest's.

Thus, Random Forest provided the best overall performance for this experiment.

22. Limitations

There are several limitations to this analysis.

First, the dataset contains only 569 observations, so larger datasets may produce different model behavior.

Second, the test-set performance depends on the particular train-test split, although stratification and a fixed random state were used.

Third, the dataset contains correlated measurements because several variables describe related characteristics such as radius, perimeter, and area.

Finally, this analysis is intended as a machine-learning experiment and should not be interpreted as a clinical diagnostic system. A real-world medical application would require additional validation, clinical evaluation, and testing on independent populations.

23. Conclusion

This project evaluated four tree-based classification algorithms on the Breast Cancer Wisconsin Diagnostic dataset: Decision Tree, Random Forest, AdaBoost, and XGBoost.

The data was first examined for missing values, duplicate observations, distributions, and correlations. The empty Unnamed: 32 column and identifier id column were removed. The categorical diagnosis variable was converted into a binary numerical target.

Standardization and normalization were performed, and eight important predictors were selected using training-data correlations.

Five-fold stratified cross-validation and GridSearchCV were used to tune the hyperparameters of each model. The models were then evaluated on an independent 20% test set using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrices.

All four models performed strongly. Random Forest produced the best overall results, achieving **95.61% accuracy and a 93.67% F1-score**, with 100% precision and 99.04% ROC-AUC. AdaBoost achieved the highest malignant recall at 90.48%, while the Decision Tree achieved the highest ROC-AUC and PR-AUC in this particular test set.

Overall, **Random Forest was selected as the best model** because it provided the strongest balance of accuracy, precision, recall, and F1-score among the four methods evaluated.